

Navid C. Constantinou

Senior Lecturer

Chief Investigator at

ARC Centre of Excellence for 21st Century Weather
School of Geography, Earth and Atmospheric Sciences
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Interests

geophysical fluid dynamics, physical oceanography, atmospheric dynamics, climate, climate modeling, machine learning, fluid mechanics, story telling (both science-related or not)

Education

- Oct. 2010 – Feb. 2015 **Ph.D. in Physics**
National & Kapodistrian University of Athens, Greece
SUPERVISOR : Petros J. Ioannou
THESIS : Formation of large-scale structures by turbulence in rotating planets [\[arXiv\]](#) 
- Sep. 2008 – Jun. 2010 **M.Sc. in Physics w/ Honours** (summa cum laude, 9.66/10)
Astrophysics, Astronomy and Mechanics
National & Kapodistrian University of Athens, Greece
- Sep. 2003 – Jun. 2008 **B.Sc. in Physics w/ Honours** (summa cum laude, class of 2008 valedictorian, 9.16/10)
National & Kapodistrian University of Athens, Greece
Exchange through Socrates-Erasmus program during spring semester 2006 at the
Rheinische Friedrich-Wilhelms Universität, Bonn, Germany
- Jul. 2001 – Aug. 2003 **Cyprus National Guard**
Military service (obligatory) as Second Lieutenant in Armored Forces, Cyprus

Appointments

- June 2024 – **Senior Lecturer**
School of Geography, Earth and Atmospheric Sciences, University of Melbourne, Australia
- June 2021 – May 2024 **ARC Discovery Early Career Researcher Award (DECRA) Fellow**
Research School of Earth Sciences, Australian National University, Australia
- May 2018 – May 2021 **Research Fellow**, part of the ARC Centre of Excellence for Climate Extremes
Research School of Earth Sciences, Australian National University, Australia (with Andy Hogg)
- Sep. 2015 – Apr. 2018 **Postdoctoral Researcher (NOAA Climate & Global Change Postdoctoral Fellow)**
Scripps Institution of Oceanography, University of California San Diego, USA (with William R. Young)
- Jun. 2015 – Aug. 2015 **Visiting Researcher**
Cyprus Oceanography Center, University of Cyprus, Cyprus

Grants

- 2024-2031 Jacob, C., Abram, N., Lane, T., Perkins-Kirkpatrick, S., Holbrook, N., Sherwood, S., Singh, M., Ritchie-Tyo, E., Arblaster, J., McGregor, S., Gallant, A., Vincent, C., King, A., Brown, J., Hogg, A., Evans, J., Alexander, L., Taschetto, A., Spence, P., Franklin, C., Pepler, A., Lim, E.-P., Brunet, G., Marsland, S., Brown, J., Risbey, J., Ziehn, T., Petch, J., Shipway, B., Senior, C., Wernli, H., Seneviratne, S., Hohenegger, C., Stevens, B., Sobel, A., Mullendore, G., Meehl, G., O’Gorman, P., Hallberg, R., Gray, S., Nazarian, N., Maher, N., **Constantinou, N. C.**, Huang, Y., Bain, C., and Dunstone, N.
ARC Centre of Excellence for the Weather of the 21st Century

ARC Centre of Excellence 2023 (ARC: 35,753,656 AUD)

- 2024-2027 **Constantinou, N. C.** (lead CI), Maher, N., and Hogg, A. McC.
An ensemble approach to studying the ocean's role in climate change
ARC Discovery Project 2024 (ARC: 504,045 AUD)
- 2022 **Constantinou, N. C.** (lead CI), Qu, Z., Hole, M., and Anzecot, O.
AI generation of cyclone data via learned physical constraints
ANU Institute for Climate, Energy & Disaster Solutions (13,500 AUD)
- 2021-2024 **Constantinou, N. C.** (sole CI)
Machine learning of subgrid ocean physics for global ocean models
ARC Discovery Early Researcher Career Award 2021 (ARC: 457,900 AUD + ANU: 126,538 AUD)
- 2015-2017 **Constantinou, N. C.** (sole CI)
NOAA Climate & Global Change Postdoctoral Fellowship (220,000 AUD)

Awards

- 2022 Most Selfless Contributor Award 2022
Consortium for Ocean and Sea Ice Modelling in Australia [\[PHOTO\]](#)
- 2019 *Best paper among Early Career Researchers within ARC Centre of Excellence for Climate Extremes*
- 2009-2014 *Alexander S. Onassis Foundation*
Scholarship for the 2nd year of M.Sc. and for 4 years of Ph.D. studies (70,000 AUD)
- 2009-2012 *A. G. Leventis Foundation*
Scholarship for the 2nd year of M.Sc. and the first 2 years of Ph.D. studies (23,500 AUD)
- 2003-2008 *Department of Physics, National & Kapodistrian University of Athens, Greece*
Valedictorian for class 2008
1st student for academic years 2003-04 and 2004-05
Honorary Scholarship for academic year 2005-06
- 2001 *International Physics Olympiad, June 2001*
Participation with the national team of Cyprus

Publications

Submitted / In Review

40. Silvestri, S., Campin, J.-M., Wagner, G. L., **Constantinou, N. C.**, Lee, X. K., and Ferrari, R. (2026). A low-storage Runge-Kutta framework for nonlinear free-surface ocean models. *J. Adv. Model. Earth Sy.* (submitted, Apr 2026; doi:[10.22541/essoar.15002225/v1](https://doi.org/10.22541/essoar.15002225/v1)) [doi](#) [download](#)
39. Bhagtani, D., Hogg, A. McC., Holmes, R. M., and **Constantinou, N. C.** (2025). Unravelling how winds and surface heat fluxes control the Atlantic Ocean's meridional heat transport. *Clim. Dyn.* (submitted, May 2025; arXiv:[2401.14230](https://arxiv.org/abs/2401.14230)) [doi](#) [download](#)
38. Wagner, G. L., Silvestri, S., **Constantinou, N. C.**, Ramadhan, A., Campin, J.-M., Hill, C., Chor, T., Strong-Wright, J., Lee, X. K., Poulin, F., Marshall, J., and Ferrari, R. (2025). High-level, high-resolution ocean modeling at all scales with Oceananigans. *J. Adv. Model. Earth Sy.* (submitted, Feb 2025; arXiv:[2502.14148](https://arxiv.org/abs/2502.14148)) [doi](#) [download](#)
37. **Constantinou, N. C.**, Rocha, C. B., Llewellyn Smith, S. G., and Young, W. R. (2023). Nusselt number scaling in horizontal convection: boundary conditions and dimensionality. *J. Fluid Mech.* (submitted, Jan 2023; revised, Jun 2023; arXiv:[2301.03122](https://arxiv.org/abs/2301.03122)) [doi](#) [download](#)
36. Wagner, G. L., **Constantinou, N. C.**, and Reichl, B. G. (2023). Stokes drift should not be added to ocean general circulation model velocities. *Geophys. Res. Lett.* (submitted Oct 2022; revised Apr 2023; arXiv:[2210.08552](https://arxiv.org/abs/2210.08552)) [doi](#) [download](#)

Published / In Press

- 2026 35. Johnston, D. R., Shakespeare, C. J., and **Constantinou, N. C.** (2026). Evaluating and improving wave and non-wave stress parameterizations for oceanic flows. *J. Phys. Oceanogr.* (in press; doi:10.1175/JPO-D-25-0064.1) [datasets and notebooks] doi 
34. Wagner, G. L. and **Constantinou, N. C.** (2025). Phenomenology of decaying turbulence beneath surface waves. *J. Fluid Mech.*, **1020**, A51. [datasets and notebooks] doi 
33. Barnes, A., Shakespeare, C., Hogg, A. McC., and **Constantinou, N. C.** (2025). Rapid topographic scatter of near-inertial waves generated by storms. *J. Phys. Oceanogr.*, **55**(10), 1847-1858. [datasets and notebooks] doi 
32. Bhagtani, D., Hogg, A. McC., Holmes, R. M., **Constantinou, N. C.**, and Khatri, H. (2025). Asymmetric response of the North Atlantic gyres to the North Atlantic Oscillation. *J. Geophys. Res. Oceans*, **130**, e2024JC021997. doi 
31. Ong, E. Q. Y., England, M. H., Doddridge, E. W., and **Constantinou, N. C.** (2025). Transient Antarctic Slope Current response to climate change including meltwater. *Geophys. Res. Lett.*, **52**, e2024GL113983. doi 
30. Silvestri, S., Wagner, G. L., **Constantinou, N. C.**, Hill, C., Campin, J.-M., Souza, A., Bishnu, S., Churavy, V., Marshall, J., and Ferrari, R. (2025). A GPU-based ocean dynamical core for routine mesoscale-resolving climate simulations. *J. Adv. Model. Earth Sy.*, **17**, e2024MS004465. doi 
- 2025 29. Wagner, G. L., Hillier, A., **Constantinou, N. C.**, Silvestri, S., Souza, A., Burns, K., Hill, C. N., Campin, J.-M., Marshall, J., and Ferrari, R. (2025). Formulation and calibration of CATKE, a one-equation parameterization for microscale ocean mixing. *J. Adv. Model. Earth Sy.*, **17**, e2024MS004522. doi 
28. Barnes, A. J., **Constantinou, N. C.**, Gibson, A., Kiss, A. E., Chapman, C., Reilly, J., Bhagtani, D., and Yang, L. (2024). regional-mom6: A Python package for automatic generation of regional configurations for the Modular Ocean Model 6. *J. Open Source Softw.*, **9**(100), 6857. (code repository; package documentation) doi 
27. Klöwer, M., Gelbrecht, M., Hotta, D., Silvestri, S., Wagner, G. L., White, A., Hatfield, S., Meyer, D., Kimpson, T., **Constantinou, N. C.**, and Hill, C. (2024). SpeedyWeather.jl: Reinventing atmospheric general circulation models towards interactivity and extensibility. *J. Open Source Softw.*, **9**(98), 6323. (code repository; package documentation) doi 
26. Bennetts, L. G., Shakespeare, C. J., Vreugdenhil, C. A., Foppert, A., Gayen, B., Meyer, A., Morrison, A. K., Padman, L., Phillips, H. E., Stevens, C. L., Toffoli, A., **Constantinou, N. C.**, Cusack, J., Cyriak, A., Doddridge, E. W., Domingues, C. M., England, M. H., Evans, D. G., Heil, P., Hogg, A. McC., Holmes, R. M., Huneke, W. G. C., Jones, N. L., Keating, S. R., Kiss, A. E., Kraitzman, N., Malyarenko, A., McConnochie, C. D., Meucci, A., Montiel, F., Neme, J., Nikurashin, M., Patel, R. S., Peng, J.-P., Rayson, M., Rosevear, M. G., Sohail, T., Spence, P., Stanley, G. J. (2024). Closing the loops on Southern Ocean dynamics: From the circumpolar current to ice shelves and from bottom mixing to surface waves. *Rev. Geophys.*, **62**, e2022RG000781. doi 
25. Silvestri, S., Wagner, G. L., Campin, J.-M., **Constantinou, N. C.**, Hill, C., Souza, A., and Ferrari, R. (2024). A new WENO-based momentum advection scheme for simulations of ocean mesoscale turbulence. *J. Adv. Model. Earth Sy.*, **16**(7), e2023MS004130. doi 
- 2024 24. Ong, E. Q. Y., Doddridge, E. W., **Constantinou, N. C.**, Hogg, A. McC., and England, M. H. (2024). Intrinsically episodic Antarctic shelf intrusions of circumpolar deep water via canyons. *J. Phys. Oceanogr.*, **54**(5), 1195-1210. doi 
23. Strong-Wright, J., Chen, S., **Constantinou, N. C.**, Silvestri, S., Wagner, G. L., and Taylor, J. R. (2023). Ocean-BioME.jl: A flexible environment for modelling the coupled interactions between ocean biogeochemistry and physics. *J. Open Source Softw.*, **8**(90), 5669. (code repository; package documentation) doi 
- 2023 22. Bhagtani, D., Hogg, A. McC., Holmes, R. M., and **Constantinou, N. C.** (2023). Surface heating steers planetary-scale ocean circulation. *J. Phys. Oceanogr.*, **53**(10), 2375-2391. doi 
21. Hogg, A. McC., Penduff, T., Close, S. E., Dewar, W. K., **Constantinou, N. C.**, and Martínez-Moreno, J. (2022). Circumpolar variations in the chaotic nature of Southern Ocean eddy dynamics. *J. Geophys. Res.-Oceans*, **127**, e2022JC018440. doi 

- 2022 20. Wagner, T. J. W., Eisenman, I., Ceroli, A. M., and **Constantinou, N. C.** (2022). How winds and ocean currents influence the drift of floating objects. *J. Phys. Oceanogr.*, **52**(5), 907-916. [doi](#) [download](#)
19. **Constantinou, N. C.** and Hogg, A. McC. (2021). Intrinsic oceanic decadal variability of upper-ocean heat content. *J. Climate*, **34**(15), 6175-6189. [[datasets and notebooks](#)] [doi](#) [download](#) (featured in the [CLEX](#) press news)
18. Martínez-Moreno, J., Hogg, A. McC., England, M. H., **Constantinou, N. C.**, Kiss, A. E., and Morrison, A. K. (2021). Global changes in oceanic mesoscale currents over the satellite altimetry record. *Nat. Clim. Chang.*, **11**, 397-403. [doi](#) [download](#) (featured in the [CLEX](#) press news; also read about it in [The Conversation](#))
Selection of Press Coverage: [The Guardian](#), [The Sydney Morning Herald](#), [Cosmos Magazine](#).
17. **Constantinou, N. C.**, Wagner, G. L., Siegelman, L., Pearson, B. C., and Palóczy, A. (2021). GeophysicalFlows.jl: Solvers for geophysical fluid dynamics problems in periodic domains on CPUs & GPUs. *J. Open Source Softw.*, **6**(60), 3053. ([code repository](#); [package documentation](#)) [doi](#) [download](#)
- 2021 16. Lozano-Durán, A., **Constantinou, N. C.**, Nikolaidis, M.-A., and Karp, M. (2021). Cause-and-effect of linear mechanisms in wall turbulence. *J. Fluid Mech.*, **914**, A8. [doi](#) [download](#)
15. Lozano-Durán, A., Nikolaidis, M.-A., **Constantinou, N. C.**, and Karp, M. (2020). Alternative physics to understand wall turbulence: Navier–Stokes equations with modified linear dynamics. *J. Phys.: Conf. Ser.*, **1522**, 012003. [doi](#) [download](#)
- 2020 14. Rocha, C. B., **Constantinou, N. C.**, Llewellyn Smith, S. G., and Young, W. R. (2020). The Nusselt numbers of horizontal convection. *J. Fluid Mech.*, **894**, A24. [doi](#) [download](#)
13. **Constantinou, N. C.** and Hogg, A. McC. (2019). Eddy saturation of the Southern Ocean: a baroclinic versus barotropic perspective. *Geophys. Res. Lett.*, **46**, 12202–12212. [[datasets and notebooks](#)] [doi](#) [download](#) (Best Early Career Researcher paper within [CLEx](#) for year 2019.)
12. Martínez-Moreno, J., Hogg, A. McC., Kiss, A. E., **Constantinou, N. C.**, and Morrison, A. K. (2019). Kinetic energy of eddy-like features from sea surface altimetry. *J. Adv. Model. Earth Sy.*, **11**(10), 3090-3105. [doi](#) [download](#) (Featured in the [CLEx](#) press news.)
11. Parker, J. B. and **Constantinou, N. C.** (2019). Magnetic eddy viscosity of mean shear flows in two-dimensional magnetohydrodynamics. *Phys. Rev. Fluids*, **4**, 083701. [doi](#) [download](#) (Featured in the [ANU](#) and [LLNL](#) press news.)
- 2019 10. Bakas, N. A., **Constantinou, N. C.**, and Ioannou, P. J. (2019). Statistical state dynamics of weak jets in barotropic beta-plane turbulence. *J. Atmos. Sci.*, **76**(3), 919-945. [doi](#) [download](#) (Featured in the [CLEx](#) press news.)
9. **Constantinou, N. C.** and Parker, J. B. (2018). Magnetic suppression of zonal flows on a beta-plane. *Astrophys. J.*, **863**, 46. [doi](#) [download](#) (Featured in the [ANU](#) and [LLNL](#) press news; also read about it in [The Conversation](#).)
- 2018 8. **Constantinou, N. C.** (2018). A barotropic model of eddy saturation. *J. Phys. Oceanogr.*, **48**(2), 397-411. [doi](#) [download](#)
- 2017 7. **Constantinou, N. C.** and Young, W. R. (2017). Beta-plane turbulence above monoscale topography. *J. Fluid Mech.*, **827**, 415-447. [doi](#) [download](#)
6. Farrell, B. F., Ioannou, P. J., Jiménez, J., **Constantinou, N. C.**, Lozano-Durán, A., and Nikolaidis, M.-A. (2016). A statistical state dynamics-based study of the structure and mechanism of large-scale motions in plane Poiseuille flow. *J. Fluid Mech.*, **809**, 290-315. [doi](#) [download](#)
- 2016 5. **Constantinou, N. C.**, Farrell, B. F., and Ioannou, P. J. (2016). Statistical state dynamics of jet–wave coexistence in barotropic beta-plane turbulence. *J. Atmos. Sci.*, **73**(5), 2229-2253. [doi](#) [download](#)
- up to 2015 4. Bakas, N. A., **Constantinou, N. C.**, and Ioannou, P. J. (2015). S3T stability of the homogeneous state of barotropic beta-plane turbulence. *J. Atmos. Sci.*, **72**(5), 1689-1712. [doi](#) [download](#)
3. **Constantinou, N. C.**, Lozano-Durán, A., Nikolaidis, M.-A., Farrell, B. F., Ioannou, P. J., and Jiménez, J. (2014). Turbulence in the highly restricted dynamics of a closure at second order: comparison with DNS. *J. Phys.: Conf. Ser.*, **506**, 012004. [doi](#) [download](#)
2. **Constantinou, N. C.**, Farrell, B. F., and Ioannou, P. J. (2014). Emergence and equilibration of jets in beta-plane turbulence: applications of Stochastic Structural Stability Theory. *J. Atmos. Sci.*, **71**(5), 1818-1842. [doi](#) [download](#)

1. **Constantinou, N. C.** and Ioannou, P. J. (2011). Optimal excitation of two dimensional Holmboe instabilities. *Phys. Fluids*, **23**, 074102. [doi](#) [download](#)

Chapters in Books (refereed)

2. **Constantinou, N. C.**, Ioannou, P. J., and Bakas, N. A. (2016). Structure and stability of low amplitude jet equilibria in barotropic turbulence. In Karacostas, T., Bais, A., and Nastos, T. P. (eds.) *Perspectives on Atmospheric Sciences*, 369-375, Springer International Publishing. [doi](#) [download](#)
- 2016 1. Bakas, N. A., **Constantinou, N. C.**, and Ioannou, P. J. (2016). On the dynamics underlying the emergence of coherent structures in barotropic turbulence. In Karacostas, T., Bais, A., and Nastos, T. P. (eds.) *Perspectives on Atmospheric Sciences*, 361-367, Springer International Publishing. [doi](#) [download](#)

Conference Proceedings (refereed)

3. Ioannou, P. J., Nikolaidis, M.-A., and **Constantinou, N. C.** (2014) Simplified turbulence in wall-bounded flows. *9th Panhellenic Meeting "Fluid Flow Phenomena" (ROH 2014)*, Athens, 12-13 Dec., 2014 (in greek). [download](#)
2. Bakas, N. A., Ioannou P. J., and **Constantinou, N. C.** (2014). Emergence of non-zonal coherent structures in barotropic turbulence. In Kanakidou, M., Mihalopoulos, N. and Nastos, P. (eds.) *Proceedings of the 12th International Conference on Meteorology, Climatology & Atmospheric Physics (COMECAP)*, Heraklion, Crete, 28-31 May, Vol. 1, 107-111, ISBN: 978-960-524-430-9. [download](#)
- 2014 1. **Constantinou, N. C.** and Ioannou, P. J. (2014). Emergence and equilibration of zonal winds in turbulent planetary atmospheres. In Kanakidou, M., Mihalopoulos, N. and Nastos, P. (eds.) *Proceedings of the 12th International Conference on Meteorology, Climatology & Atmospheric Physics (COMECAP)*, Heraklion, Crete, 28-31 May, Vol. 1, 210-214, ISBN: 978-960-524-430-9. [download](#)

Other publications

- 2023 7. Silvestri, S., Wagner, G. L., Hill, C., Ardakani, M. R., Blaschke, J., Campin, J.-M., Churavy, V., **Constantinou, N. C.**, Edelman, A., Marshall, J., Ramadhan, A., Souza, A., Ferrari, R. (2023). Oceananigans.jl: A model that achieves breakthrough resolution, memory and energy efficiency in global ocean simulations. (arXiv:2309.06662) [doi](#) [download](#)
- 2022 6. Miller, J. W., O'Neil, C., **Constantinou, N. C.**, and Anzecot, O. (2022). Eigenvalue initialisation and regularisation for Koopman autoencoders. (arXiv:2212.12086) [doi](#) [download](#)
5. **Constantinou, N. C.** (2021). How machine learning is helping us fine-tune climate models to reach unprecedented detail. *The Conversation*, 18th August 2021. [URL](#)
- 2021 4. **Constantinou, N. C.**, Martínez-Moreno, J., Hogg, A. McC., England, M. H., Kiss, A. E., and Morrison, A. K. (2021). Satellites reveal ocean currents are getting stronger, with potentially significant implications for climate change. *The Conversation*, 23rd April 2021. [URL](#)
- 2019 3. Lozano-Durán, A., Nikolaidis, M.-A., **Constantinou, N. C.**, and Karp, M. (2019). Wall turbulence without modal instability of the streaks. (arXiv:1909.05490) [download](#)
2. Lozano-Durán, A., Karp, M., and **Constantinou, N. C.** (2018). Wall turbulence with constrained energy extraction from the mean flow. *Center for Turbulence Research – Annual Research Briefs 2018*, 209-220. [download](#)
- 2018 1. **Constantinou, N. C.** (2018). Jupiter's magnetic fields may stop its wind bands from going deep into the gas giant. *The Conversation*, 10th August 2018. [URL](#)

Conferences

- 2023 Oceananigans.jl: breakthrough resolution, memory, and energy efficiency in global ocean simulations. *Australian Leadership Computing Symposium 2023* [\[URL\]](#), Australian Academy of Sciences, Canberra, Australia, 14-16 Jun. 2023. (talk) [↓](#)
- Open-source, reproducible workflow in physical oceanography and geophysical fluid dynamics. *Ocean Sciences Meeting 2022*, Virtual Conference, USA, 24 Feb.-4 Mar 2022. **(invited talk)**
- 2022 A data-driven approach for developing and calibrating a parameterization for mesoscale eddy fluxes. *Ocean Sciences Meeting 2022*, Virtual Conference, USA, 24 Feb.-4 Mar 2022. (talk)
- 2021 A data-driven approach for developing and calibrating a parameterization of the ocean mesoscale eddy fluxes *Conference on “Machine Learning for Climate”*, KITP, UC Santa Barbara, USA, 1-4 Nov. 2021. **(invited talk)** [↓](#) [\[VIDEO\]](#)
- Cause-and-effect of linear mechanisms in wall turbulence. *73rd APS Division of Fluid Dynamics Meeting*, Chicago, IL, USA, 22-24 Nov. 2020. (virtual talk) [↓](#) [\[VIDEO\]](#)
- 2020 Eddy saturation of the Southern Ocean: a baroclinic versus barotropic perspective. *Ocean Sciences Meeting 2020*, San Diego, CA, USA, 16-21 Feb. 2020. (poster) [↓](#)
- Demystifying the Southern Ocean’s response to wind variability. *ARC Centre of Excellence for Climate Extremes Annual Workshop 2019* [\[URL\]](#), Hobart, Tasmania, Australia, 19-21 Nov. 2019. **(invited talk)** [↓](#)
- Barotropic versus baroclinic eddy saturation: implications to Southern Ocean dynamics. *22nd Conference on Atmospheric and Oceanic Fluid Dynamics*, Portland ME, USA, 24-28 Jun. 2019. (talk) [↓](#)
- 2019 Magnetic eddy viscosity of mean shear flows in 2D magnetohydrodynamics: possible application to gas giants’ interiors. *22nd Conference on Atmospheric and Oceanic Fluid Dynamics*, Portland ME, USA, 24-28 Jun. 2019. (poster) [↓](#)
- Barotropic versus baroclinic eddy saturation. *AGU Fall Meeting 2018*, Washington DC, USA, 10-14 Dec. 2018. (poster) [↓](#)
- Magnetic suppression of zonal flows on a beta plane. *AGU Fall Meeting 2018*, Washington DC, USA, 10-14 Dec. 2018. (poster) [↓](#)
- 2019 Statistical state dynamics reveals mechanism for organization of coherent structures in turbulent flows. *Eurochem Colloquium 598: Coherent structures in wall-bounded turbulence*, Imperial College London, London, UK, 29-31 Aug. 2018. **(invited keynote talk)** [↓](#)
- Eddy saturation in a barotropic model. *21st Conference on Atmospheric and Oceanic Fluid Dynamics*, Portland OR, USA, 25-30 Jun. 2017. (talk) [↓](#)
- A statistical state dynamics based theory for jet–wave coexistence in beta-plane turbulence. *21st Conference on Atmospheric and Oceanic Fluid Dynamics*, Portland OR, USA, 25-30 Jun. 2017. (poster) [↓](#)
- 2017 Understanding self-organization in turbulent flows by studying the statistical state dynamics, *Conference on “Recurrence, self-organization, and the dynamics of turbulence”*, KITP, UC Santa Barbara, USA, 9-13 Jan. 2017. **(invited talk)** [↓](#)
- Topographic beta-plane turbulence and form stress. *AGU Fall Meeting 2016*, San Francisco, USA, 12-16 Dec. 2016. (poster) [↓](#)
- Structure and mechanism of turbulence under dynamical restriction in plane Poiseuille flow. *69th APS Division of Fluid Dynamics Meeting*, Portland, USA, 20-22 Nov. 2016. (talk) [↓](#)
- 2016 Statistical state dynamics of jet–wave coexistence in beta-plane turbulence. *APS March Meeting 2016*, Baltimore, USA, 14-18 Mar., 2016. (talk) [↓](#)
- up to 2014 Emergence and equilibration of zonal winds in turbulent planetary atmospheres. *12th International Conference on Meteorology, Climatology and Atmospheric Physics, COMECAP 2014* [\[URL\]](#), Heraklion, Crete, Greece, 28-31 May 2014. (poster) [↓](#)
- Emergence and equilibration of jets in planetary turbulence. *EGU 2013 General Assembly* [\[URL\]](#), Vienna, Austria, 8-12 Apr. 2013. (talk) [↓](#)

Emergence and equilibration of jets in planetary turbulence. *8th Panhellenic Meeting “Fluid Flow Phenomena” (ROI 2012)* [\[URL\]](#), Volos, Greece, 16-17 November 2012. (talk) [↓](#)

Seminars

“From little things, big things grow”: How small-scale ocean turbulence affects the global ocean circulation and climate. School of Geography, Earth and Atmospheric Sciences [\[URL\]](#), University of Melbourne [\[URL\]](#), Melbourne, Australia, 15 Nov. 2023. **(invited talk)**

2023 Open-source and reproducibility workflow within the Consortium for Ocean–Sea Ice Modelling in Australia (COSIMA). Pangeo Showcase [\[URL\]](#) (via Zoom), 10 May 2023. **(invited talk)**

2022 “From little things big things grow”: how mesoscale eddies affect the global ocean circulation and climate. Scripps Institution of Oceanography, UC San Diego [\[URL\]](#), La Jolla, CA, USA, 12 May 2022. **(invited talk)**

2021 From small swirls up to the global ocean circulation: how ocean eddies affect the Earth’s climate, Research School of Earth Sciences [\[URL\]](#), Australian National University, Canberra, Australia, 25 Mar. 2021. **(invited talk)** [↓](#) [\[VIDEO\]](#)

2020 Cause-and-effect of linear mechanisms in wall turbulence. Shear Flow Instability, Transition and Turbulence Seminar Series, Monash University [\[URL\]](#), (via Zoom), 7 Oct. 2020. [↓](#)

What’s underneath Jupiter’s and Saturn’s stripes? FEARS Meeting, Research School of Astronomy and Astrophysics [\[URL\]](#), Australian National University, Canberra, Australia, 29 Oct. 2019. [↓](#)

What is hiding underneath the stripes of Jupiter and Saturn? Department of Physics Colloquium, Department of Physics [\[URL\]](#), National and Kapodistrian University of Athens [\[URL\]](#), Athens, Greece, 16 Oct. 2019. [↓](#)

Barotropic versus baroclinic eddy saturation: implications to Southern Ocean dynamics. Geophysical Fluid Dynamics Summer Program, Woods Hole Oceanographic Institution [\[URL\]](#), Woods Hole, USA, 9 Jul. 2019. [↓](#)

Barotropic versus baroclinic eddy saturation: implications to Southern Ocean dynamics. Geophysical Fluid Dynamics Summer Program, Woods Hole Oceanographic Institution [\[URL\]](#), Woods Hole, USA, 9 Jul. 2019. [↓](#)

A barotropic process-model for eddy saturation. WHOI Physical Oceanography Seminar Series, Woods Hole Oceanographic Institution [\[URL\]](#), Woods Hole, USA, 16 Apr. 2019. [↓](#)

How does the Antarctic Circumpolar Current respond to the increasing winds over the Southern Ocean? Barotropic versus baroclinic eddy saturation. Physics & Physical Oceanography Department Seminar Series, University of North Carolina Wilmington [\[URL\]](#), Wilmington NC, USA, 11 Apr. 2019.

How does the Antarctic Circumpolar Current respond to the increasing winds over the Southern Ocean?: Barotropic versus Baroclinic Eddy Saturation. Fluids Seminar Math Department [\[URL\]](#), Monash University [\[URL\]](#), Melbourne, Australia, 12 Feb. 2019.

2019 Statistical state dynamics: a new framework for understanding turbulent flows. Fluid Mechanics Research Group Seminar [\[URL\]](#), University of Melbourne [\[URL\]](#), Melbourne, Australia, 8 Feb. 2019.

Magnetic suppression of zonal flows on a beta plane. SpinLab Group Seminar [\[URL\]](#), University of California Los Angeles [\[URL\]](#), Los Angeles, CA, USA, 11 Jan. 2019. [↓](#)

2018 Barotropic versus baroclinic eddy saturation. Oceans and Climate Group Seminar [\[URL\]](#), Geophysical Fluid Dynamics Laboratory [\[URL\]](#), Princeton, NJ, USA, 12 Dec. 2018. [↓](#)

Eddy saturation in a barotropic model. LDEO OCP Seminar [\[URL\]](#), Lamont-Doherty Earth Observatory [\[URL\]](#), Columbia University, Palisades, NY, USA, 27 Oct. 2017. [↓](#)

Eddy saturation in a barotropic model. ClimaTea Seminar [\[URL\]](#), Harvard University, Cambridge, MA, USA, 26 Oct. 2017. [↓](#)

2017 Eddy saturation in a barotropic model. CEAFM Seminar [\[URL\]](#), Department of Earth & Planetary Sciences, The Johns Hopkins University, Baltimore, USA, 13 Oct. 2017. [↓](#)

Topographic beta-plane turbulence and form stress. Geophysical Fluid Dynamics Summer Program, Woods Hole Oceanographic Institution [\[URL\]](#), Woods Hole, USA, 19 Jul. 2016. [↓](#)

Topographic beta-plane turbulence and form stress. Mathematics of Turbulence Reunion Conference, IPAM, UCLA [\[URL\]](#), Lake Arrowhead, 7 Jun. 2016. [↓](#)

Statistical state dynamics of planetary turbulence. CEA-FM Seminar [\[URL\]](#), Whiting School of Engineering, The Johns Hopkins University, Baltimore, 18 Mar. 2016. [↓](#)

2016 A theory for large-scale structure formation in atmospheric/oceanic turbulence: Is jet formation a phase transition phenomenon? CASPO Seminar, Scripps Institution of Oceanography, UC San Diego [\[URL\]](#), La Jolla, 10 Feb. 2016. [↓](#)

up to 2015 Formation of large-scale structures by turbulence in planetary atmospheres. Physics Department, University of Cyprus [\[URL\]](#), Nicosia, 5 May 2015. [↓](#)

Emergence of large-scale structure in planetary turbulence as an instability of the homogeneous turbulent state. IPAM, UCLA [\[URL\]](#), Los Angeles, 21 Oct. 2014. [↓](#)

Emergence and equilibration of zonal winds in turbulent planetary atmospheres. Cyprus Oceanography Center, University of Cyprus [\[URL\]](#), Nicosia, 7 Jan. 2014. [↓](#)

Verification of the predictions of SSST in nonlinear simulations. 2nd Meeting of “Zonal Jets and Eddies” team, International Space Science Institute (ISSI) [\[URL\]](#), Bern 2-5 Apr. 2013. [↓](#)

Workshops

2023, 2024 **Sean R. Haney Memorial Symposium** (organizer)
Scripps Institution of Oceanography, US San Diego
February 2023, 2024, La Jolla, CA, USA

2021 **Machine Learning and the Physics of Climate**
Kavli Institute for Theoretical Physics, UC Santa Barbara
November 1 - December 17 2021, Santa Barbara, CA, USA [\[URL\]](#)

2021 **Layering in Atmospheres, Oceans, and Plasmas**
Kavli Institute for Theoretical Physics, UC Santa Barbara
January 11 - March 12 2021, Santa Barbara (Virtual), CA, USA [\[URL\]](#)

2019 **Advanced Ocean Modelling School**
University of Tasmania
April 28 - May 3, Lake Pedder, Tasmania, Australia [\[URL\]](#)

2017 **Vorticity in the Universe: From Superfluids to Weather and Climate, to the Universe**
Aspen Center for Physics
August 27 - September 17 2017, Aspen, CO, USA [\[URL\]](#)

2017 **Les Houches Summer School on Fundamental Aspects of Turbulent Flows in Climate Dynamics**
Les Houches Physics School
July 31 - August 25 2017, Les Houches, France [\[URL\]](#)

2014 **Mathematics of Turbulence**
Institute of Pure & Applied Mathematics, UCLA
September 8 - December 12 2014, Los Angeles, USA [\[URL\]](#)

2013 **Geoturb: Numerical Modeling and Theoretical Challenges in Atmosphere and Ocean Turbulence**
Ecole normale supérieure de Lyon
2-4 October 2013, Lyon, France. [\[URL\]](#)

2013 **First Multiflow Summer Workshop**
Universidad Politécnica de Madrid
10 June - 12 July 2013, Madrid, Spain. [\[URL\]](#)

2011 **International Graduate School on Stability, Transition to Turbulence and Flow Control**
Organized by Advanced Instability Methods (AIM) Network
22-27 August 2011, Cambridge, UK. [\[URL\]](#)

2009 **Climate Variability & Climate Change: Estimating and Reducing Uncertainties**
8-17 June 2009, Visegrád, Hungary. [\[URL\]](#)

Teaching

2024 – **Climate Dynamics** (3rd year Undergraduate EMSC3039)
(along w/ Callum Shakespeare (convener), Kial Stewart, and Nicola Maher)
Research School of Earth Sciences, Australian National University, Australia

2021 – 2023 **Computational Geosciences** (Undergraduate/Honours EMSC4033; Masters EMSC8033)
(main lecturer and co-convener along w/ Louis Moresi)
[course website at [github](#)]
Research School of Earth Sciences, Australian National University, Australia

Jun. 2021 **Atmosphere and Ocean Dynamics Winter School 2021**
[Postponed due to COVID-19 lockdowns in East Australia]
(main lecturer along w/ Martin Singh and Annie Foppert)
Australian National University
[organized by ARC Centre of Excellence for Climate Extremes; more information [online](#)]

Jun. 2020 **Atmosphere and Ocean Dynamics Winter School 2020**
[School changed form due to COVID-19; 3 introductory lectures via Zoom, full-blown lectures postponed for 2021]
(main lecturer along w/ Martin Singh and Annie Foppert)
Institute for Marine and Antarctic Studies & University of Tasmania
[organized by ARC Centre of Excellence for Climate Extremes; more information [online](#)]

2020 Course Coordinator for **Basics of Dynamical Systems and Bifurcation Theory** (Honours/Masters/Graduate)
[3 lectures by Henk Dijkstra; 3 workshops by myself; lecture & workshop slides/notes available at [github](#)]
Research School of Earth Sciences, Australian National University, Australia

2018 Course Coordinator for **Instabilities in Fluids** (Honours/Masters)
[lecture notes and students' project reports available at [github](#)]
Research School of Earth Sciences, Australian National University, Australia

2018 Visiting lecturer for **Fluid Mechanics** (Undergraduate) [lecture notes available at [github](#)]
Department of Physics and Physical Oceanography, University of North Carolina Wilmington, USA

2017 Teaching assistant for **Applied Mathematics III** (Graduate)
Scripps Institution of Oceanography, University of California San Diego, USA

2010-2014 Teaching assistant for **Nonlinear Dynamical Systems** (3rd year Undergraduate)
Physics Department, National & Kapodistrian University of Athens, Greece

Students

Ph.D.

Mar. 2026 – Isaac Miles-Fine (School of Geography, Earth and Atmospheric Sciences, University of Melbourne; co-supervisor).

Feb. 2026 – Brodie Ellis (Victorian College of the Arts, University of Melbourne; co-supervisor).

Aug. 2025 – Waylon Wu (School of Mathematics and Statistics, University of Melbourne; co-supervisor).

Feb. 2025 – James Thompson (School of Geography, Earth and Atmospheric Sciences, University of Melbourne; co-supervisor).

Jul. 2024 – Md Ayshik Rahman Khan (School of Geography, Earth and Atmospheric Sciences, University of Melbourne; co-supervisor).

Feb. 2021 – Ellie Ong (Climate Change Research Centre, University of New South Wales; co-supervisor).
Dec. 2024 Thesis: **“Variability of the Antarctic slope and coastal currents.”**

Oct. 2020 – Dhruv Bhagtani (Research School of Earth Sciences, Australian National University; co-supervisor).
Dec. 2024 Thesis: **“Ocean gyres driven by wind stress and surface buoyancy forcing.”**

Masters/Honours

Feb. 2026 – Lara Buhagiar (MSc Earth Science, University of Melbourne).
Project: **“Open ocean convection.”** [preliminary title]
Jul. 2025 – Tatiana Prada Lopez (MSc Climate Science, University of Melbourne).
Jun. 2026 Project: **“Ocean salinity infilling from temperature observations using machine learning.”**
Feb. 2022 – Elise Palethorpe (Honours, Australian National University).
Nov. 2022 Project: **“Implementing a multigrid pressure solver in CliMA’s ocean general circulation model.”**
First Class Honours

Undergraduate

Dec. 2022 – Phoebe Grosser (2nd year, Australian National University).
Mar. 2023 Project: **“Ocean’s tidal response with high-fidelity bathymetry and investigation on physical mechanisms.”**
Jul. 2022 – Jack Miller (2nd year, Australian National University).
Dec. 2022 Project: **“Modeling with Koopman autoencoders for data synthesis with application to cyclones.”**
Feb. 2022 – Jack Miller (2nd year, Australian National University).
Jul. 2022 Project: **“Predicting cyclone genesis, trajectory, and intensity with machine learning.”**
Feb. 2022 – Oliver Balfour (2nd year, Australian National University).
Jul. 2022 Project: **“Predicting cyclone genesis, trajectory, and intensity with machine learning.”**
Jul. 2020 – Elise Palethorpe (2nd year, Australian National University).
Nov. 2020 Project: **“Numerical methods for Partial Differential Equations.”**
Aim: Learn basics of finite difference numerical schemes for solving hyperbolic and also study and implement high-order accurate weighted essentially non-oscillatory (WENO) schemes.
Nov. 2019 – Fabian Antonio Circelli (3rd year, Australian National University).
Feb. 2020 Project: **“Fourier-based Pseudospectral Methods for Solving Partial Differential Equations.”**
Aim: Learn basics of pseudospectral techniques for solving PDEs and implement GPU functionality in Fourier-Flows.jl Julia package.

Programming skills

 **julia**, Python, git

Open-Source Software

A small selection of projects include:

`Oceananigans.jl`: A fast and friendly incompressible fluid flow solver in Julia that can be run in 1-3 dimensions on CPUs and GPUs; doi:[10.21105/joss.02018](https://doi.org/10.21105/joss.02018). (**main developer**)

`FourierFlows.jl`: Julia ecosystem for solving partial differential equations on periodic domains with Fourier-based pseudospectral methods; doi:[10.5281/zenodo.1161724](https://doi.org/10.5281/zenodo.1161724). (**main developer**)

`GeophysicalFlows.jl`: Julia modules for solving problems in Geophysical Fluid Dynamics on periodic domains using Fourier-based pseudospectral methods; doi:[10.21105/joss.03053](https://doi.org/10.21105/joss.03053). (**lead developer**)

`ParameterEstimocean.jl`: Parameter estimation of turbulence closures for ocean models using Ensemble Kalman Inversion; doi:[10.5281/zenodo.5762810](https://doi.org/10.5281/zenodo.5762810). (**main developer**)

`OceanBioME.jl`: A fast and flexible modelling environment written in Julia for modelling the coupled interactions between ocean biogeochemistry, carbonate chemistry, and physics; doi:[10.21105/joss.05669](https://doi.org/10.21105/joss.05669). (**main**

developer)

xrft: Python package for discrete Fourier transforms (DFT) on xarray and dask arrays; doi:[10.5281/zenodo.1402635](https://doi.org/10.5281/zenodo.1402635).
(**contributor**)

For more details visit my GitHub profile:  [navidcy](#)

Outreach

I regularly contribute articles to [The Conversation](#) and also often appear in radio shows talking about science and been invited to give [public audience lectures](#).

A selection of outreach events is found at my website: www.navidconstantinou.com/outreach/

Mobility

Studied and worked in academic institutions in **Greece**, **Cyprus**, **U.S.A.**, and **Australia**.

Hobbies

surfing 🏄, biking 🚲, horse riding 🐎, dancing 🕺, photography 📷, story telling, and story listening

References

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